

The maximal energy of integral circulant graphs of orders $p^{2r}3^{2s+1}$ and pq^m

Horace Dong

Independent researcher

maxwelldhx@gmail.com

ORCID: [0009-0002-7554-7548](https://orcid.org/0009-0002-7554-7548)

September 2026

Abstract

Jiang and Yang determined the maximal energy of integral circulant graphs of order $n = p^{2r}q^{2s+1}$, where p, q are distinct odd primes with $q \geq 5$: the checkerboard divisor set $\{p^i q^j : i+j \text{ even}\}$ is the unique maximiser. They noted that their semidefinite method does not extend to $q = 3$. We prove the remaining case: for every prime $p \geq 5$ and all $r \geq 1$, $s \geq 0$, the checkerboard set is the unique energy maximiser for order $p^{2r}3^{2s+1}$, with the maximal energy given by the formula of Jiang and Yang. This settles the maximal-energy problem for opposite-parity exponents for all pairs of distinct odd primes. The key step is a sharp inequality for sign matrices under weighted Ramanujan transforms. We prove it with a path identity and an explicit potential function, and the same argument gives a semidefinite-free proof of the theorem of Jiang and Yang. For exponents of equal parity the checkerboard set contains n and is not admissible. We conjecture the maximal energy in this case, with two maximisers, and prove the conjecture for the orders pq^m and $p^m q$ with m odd and p, q any distinct odd primes; for $n = pq$ it also follows from known energy formulas. With the opposite-parity case, this determines the maximal energy for every order pq^m . Exact computer-assisted certificates for eleven further exponent pairs show that the conjecture holds for all such orders $p^a q^b$ with $(a+1)(b+1) \leq 27$. The main theorem, the sign-matrix inequality, the energy formula for these graphs and the theorem for the orders pq^m and $p^m q$ are formally verified in Lean 4; the closed-form evaluation of the maximal energies and the certificates are not formalised. The proofs were found by AI coding agents working under the author's direction; their role is described at the end of the paper.

1 Introduction

The energy of a finite graph is the sum of the absolute values of its adjacency eigenvalues [2]. For $n \geq 2$ and a nonempty set $\mathcal{D}$ of proper divisors of n , the integral circulant graph $\text{ICG}_n(\mathcal{D})$ has vertex set $\mathbb{Z}/n\mathbb{Z}$, two distinct vertices u, v being adjacent when $\gcd(u - v, n) \in \mathcal{D}$. These are the circulant graphs with integral spectrum [11]. The problem of maximising the energy of $\text{ICG}_n(\mathcal{D})$ over $\mathcal{D}$ for fixed n has been solved for prime powers [9] and studied for other orders; see the introduction of [6] for a survey. Roldán [8] conjectured that for $n = p^2 q^3$ the checkerboard set $\{1, p^2, pq, q^2, p^2 q^2, pq^3\}$ is the unique maximiser. Jiang and Yang [6] proved much more.

Theorem 1 (Jiang–Yang [6, Theorem A]). *Let p, q be distinct odd primes with $q \geq 5$, let $r \geq 1$, $s \geq 0$ and $n = p^{2r}q^{2s+1}$. Among all nonempty sets of proper divisors of n , the energy of $\text{ICG}_n(\mathcal{D})$ is uniquely maximised by*

$$\mathcal{D}_{r,s}^* = \{p^i q^j : 0 \leq i \leq 2r, 0 \leq j \leq 2s+1, i+j \text{ even}\},$$

and the maximal energy is $\frac{1}{2} [n + d_{2r}(p) d_{2s+1}(q)]$, where $d_k(x) = (2k+1)x^k + 4 \sum_{j=0}^{k-1} (-1)^{k-j} (j+1)x^j$.

They remark that the case $q = 3$ is not covered, because their odd-exponent semidefinite argument fails at $x = 3$ [6, Remark 4.2]. Our main result removes this restriction.

Theorem 2. *The conclusion of Theorem 1 holds for $q = 3$: for every prime $p \geq 5$ and all $r \geq 1$, $s \geq 0$, the set $\mathcal{D}_{r,s}^*$ is the unique energy maximiser among nonempty sets of proper divisors of $n = p^{2r}3^{2s+1}$, and the maximal energy is $\frac{1}{2}[n + d_{2r}(p) d_{2s+1}(3)]$.*

Together with Theorem 1 (which allows $p = 3$ on the even side), Theorem 2 determines the maximal energy and the unique maximiser for every order $p^{\text{even}}q^{\text{odd}}$ with p, q distinct odd primes. The case $r = s = 1$ was treated earlier. Park [7] claims a proof of Roldán's conjecture for all distinct odd primes. Schreib [10] gives a written proof of the case $q = 3$ and combines it with [6] to obtain the conjecture. An arithmetic certificate for $n = 27p^2$ was announced in [12]. These works concern only the exponents $(2, 3)$, and we are not aware of any earlier treatment of the case $q = 3$ for general r and s .

If the two exponents have the same parity, the checkerboard set contains n itself and is not a set of proper divisors, and the maximiser changes. For this case we state a conjecture (Conjecture 12), with two maximisers and an explicit maximal energy, and we prove it when one of the exponents equals 1. For $n = p^a q^b$, a set written as $\{p^i q^j : \dots\}$ consists of the divisors $p^i q^j$ of n ($0 \leq i \leq a$, $0 \leq j \leq b$) with the stated property, and $\delta_k(x) = (x^k(x-1) + 2(-1)^k)/(x+1)$ (Remark 7).

Theorem 3. *Let p, q be distinct odd primes, let $m \geq 1$ be odd, and let $n = pq^m$. The maximal energy of an integral circulant graph of order n is*

$$\frac{1}{2}[n + (3p - 4) d_m(q)] - (p - 2) \delta_m(q),$$

and it is attained exactly by the sets $\{p^i q^j : i + j \text{ odd}\}$ and $\{p^i q^j : i + j \text{ even}\} \setminus \{n\}$. Exchanging the names of the primes, for $n = p^m q$ the maximal energy is $\frac{1}{2}[n + (3q - 4) d_m(p)] - (q - 2) \delta_m(p)$, attained exactly by the two sets with the same description.

For $m = 1$ the maximal energy is $4(p-1)(q-1)$, attained by $\{p, q\}$ and $\{1\}$. This case can also be derived from published energy formulas of Ilić [3] and of Ilić and Bašić [4] (Remark 14), although we have not found it stated. We are not aware of an earlier determination of the maximal energy for other orders with exponents of equal parity. If m is even, the exponents of pq^m have opposite parity. Theorems 1, 2 and 3 therefore determine the maximal energy, and all maximisers, for every order pq^m ($m \geq 1$) with p, q distinct odd primes. Exact computer-assisted certificates (Proposition 13) prove the conjecture for every exponent pair (a, b) with $a, b \geq 2$ and $(a+1)(b+1) \leq 27$; in general it remains open.

Our proof replaces the semidefinite part of [6] by Theorem 10 below, an inequality for sign matrices that holds for all exponents and parities and all real $p \geq 5$, $q \geq 3$. By symmetry it also yields Theorem 1 without semidefinite certificates (Remark 11). Section 8 treats exponents of equal parity, and Section 9 proves Theorem 3; the proof applies the path identity and the potential of Sections 3 and 4 to the two rows of the sign matrix.

2 Weighted Ramanujan transforms

We use the notation of [6]. For real $x > 1$, an integer $k \geq 1$ and $0 \leq i, j \leq k$, let $T_k(x) = (t_{ij})$ be the symmetric matrix with

$$t_{ij} = \begin{cases} 0, & i + j \leq k - 2, \\ -x^{k-1}(x-1), & i + j = k - 1, \\ x^{2k-i-j-2}(x-1)^2, & i + j \geq k, \end{cases} \quad (i, j < k),$$

$t_{ik} = t_{ki} = x^{k-i-1}(x-1)$ for $i < k$, and $t_{kk} = 1$. When x is prime, $t_{ij} = \varphi(x^{k-i}) c_{x^{k-j}}(x^i)$, where $c_m(u)$ denotes Ramanujan's sum [6, (2.2)–(2.4)]. The row sums are $T_k(x)\mathbf{1} = x^k e_k$ [6, (2.5)], where e_k is the last standard basis vector. For a real matrix Z let $\|Z\|_1$ be the sum of the absolute values of its entries.

Proposition 4 ([6, Proposition 2.2]). *Let $p \neq q$ be primes, $a, b \geq 1$, $n = p^a q^b$, and let $X \in \{0, 1\}^{(a+1) \times (b+1)}$ have $x_{ij} = 1$ exactly when $p^i q^j \in \mathcal{D}$. Then $\mathcal{E}(\text{ICG}_n(\mathcal{D})) = \|T_a(p)XT_b(q)^\top\|_1$.*

Let $s_k = (1, -1, 1, \dots, (-1)^k)^\top$ and define $d_k(x) = \|T_k(x)s_k\|_1$, and put $d_0(x) = 1$. By Remark 7 below, $d_k(x)$ agrees with the closed form in Theorem 1 for all real $x > 2$; see also [6, (3.6)].

3 A path identity

For $w \in \mathbb{R}^{k+1}$ define $z_{k-1} = w_k$ and, for $j = k-1, k-2, \dots, 0$,

$$z_{j-1} = \frac{z_j + (x-1)w_j}{x}. \quad (1)$$

Thus z_j is a convex combination of $w_{j+1}, \dots, w_k$, and each z_j is a linear function of w .

Lemma 5. *For $w \in \mathbb{R}^{k+1}$ we have*

$$\|T_k(x)w\|_1 = x^k \left(|z_{-1}| + \sum_{j=0}^{k-1} |z_{j-1} - z_j| \right).$$

Proof. Unwinding (1) gives $z_j = x^{-(k-j-1)} \sum_{i>j} \varphi(x^{k-i}) w_i$ with $\varphi(x^0) = 1$ and $\varphi(x^e) = x^{e-1}(x-1)$. The last row of $T_k(x)$ is $(\varphi(x^{k-i}))_i$, so $(T_k(x)w)_k = x^k z_{-1}$. For $u < k$ put $j = k-1-u$. Row u has entry $-x^{k-1}(x-1)$ in column j , entries $x^j(x-1)\varphi(x^{k-i})$ in columns $i > j$, and zeros elsewhere. Hence $(T_k(x)w)_u = x^{k-1}(x-1)(z_j - w_j)$, while $z_{j-1} - z_j = \frac{x-1}{x}(w_j - z_j)$. $\square$

4 The one-dimensional potential

For $x > 2$ let $\mu_{-1} = 1$, $\mu_j = (x-1-\mu_{j-1})/x$ for $j \geq 0$, and $c_j = (x-1)(1+\mu_{j-1})/x$. By induction, $\mu_j \in [\frac{x-2}{x}, 1]$ for all j .

Lemma 6. *Let $x > 2$, let $y \in \{\pm 1\}^{k+1}$, and let z_j be defined by (1) with $w = y$. Then $\text{sgn } z_j = y_{j+1}$ and $\frac{x-2}{x} \leq |z_j| \leq 1$ for $-1 \leq j \leq k-1$, and*

$$\frac{\|T_k(x)y\|_1}{x^k} = \sum_{j=0}^{k-1} c_j + \mu_{k-1} - \sum_{j \in \text{NA}(y)} 2\mu_j |z_j|, \quad \text{NA}(y) = \{0 \leq j \leq k-1 : y_j = y_{j+1}\}.$$

Consequently $\|T_k(x)y\|_1 \leq d_k(x)$, with equality exactly when $y = \pm s_k$, and $d_k(x) = x^k(\sum_j c_j + \mu_{k-1})$. Moreover $(T_k(x)s_k)_k > 0$.

Proof. We have $z_{k-1} = y_k$, and since $x-1 > 1 \geq |z_j|$, the sign of $z_{j-1} = (z_j + (x-1)y_j)/x$ is y_j and $|z_{j-1}| = (x-1 \pm |z_j|)/x \in [\frac{x-2}{x}, 1]$. Put $f_j = |z_{j-1} - z_j| + \mu_{j-1}|z_{j-1}| - \mu_j|z_j|$. If $y_j = -\text{sgn } z_j$, then $|z_{j-1} - z_j| = \frac{x-1}{x}(1+|z_j|)$ and $|z_{j-1}| = \frac{x-1+|z_j|}{x}$; using $x\mu_j = x-1-\mu_{j-1}$ one finds $f_j = c_j$. If $y_j = \text{sgn } z_j$, the same computation with $|z_{j-1} - z_j| = \frac{x-1}{x}(1-|z_j|)$ and $|z_{j-1}| = \frac{x-1+|z_j|}{x}$ gives $f_j = c_j - 2\mu_j|z_j|$. Summing, the terms $\mu_j|z_j|$ telescope, and Lemma 5 gives the identity, since $\mu_{-1} = 1$ and $|z_{k-1}| = 1$. Every $j \in \text{NA}(y)$ contributes a positive term $2\mu_j|z_j|$, and $\text{NA}(y) = \emptyset$ exactly for $y = \pm s_k$. Finally $(T_k(x)s_k)_k = x^k z_{-1}$ and $\text{sgn } z_{-1} = (s_k)_0 = 1$. $\square$

Remark 7. Put $\delta_k(x) = x^k \mu_{k-1}$. Solving the recursion for μ gives $\delta_k(x) = (x^k(x-1) + 2(-1)^k)/(x+1)$. For $y = s_k$, every step in the proof of Lemma 6 has $y_j = -\text{sgn } z_j$, so $|z_{j-1}| = (x-1-|z_j|)/x$; starting from $|z_{k-1}| = 1 = \mu_{-1}$ this gives $|z_j| = \mu_{k-2-j}$, and in particular $(T_k(x)s_k)_k = x^k z_{-1} = \delta_k(x)$. Since the potential does not depend on k , Lemma 6 gives $d_{k+1}/x^{k+1} - d_k/x^k = c_k + \mu_k - \mu_{k-1} = 2\mu_k$, that is, $d_{k+1}(x) = x d_k(x) + 2\delta_{k+1}(x)$, with $d_0 = 1$. The closed form $(2k+1)x^k + 4 \sum_{j=0}^{k-1} (-1)^{k-j} (j+1)x^j$ satisfies the same recursion and initial value, so it equals $d_k(x)$ for all $k \geq 0$.

5 A two-variable inequality

Lemma 8. Let $P \geq \mu$ and $0 \leq \mu \leq 1$, and put $h(A, B) = P|A - B| + \mu|B + PA| - (P - \mu)|B| - P(1 + \mu)|A|$ for real A, B . Then $h(A, B) \leq 2\mu \max\{0, |B| - P|A|\}$.

Proof. As $h(-A, -B) = h(A, B)$ we may assume $A \geq 0$. If $0 \leq B \leq A$ then $h = -2(P - \mu)B \leq 0$. If $B > A$ then $h = 2(\mu B - PA) \leq 2\mu(B - PA)$ because $\mu \leq 1$. If $-PA \leq B < 0$ then $h = 0$. If $B < -PA$ then $h = 2\mu(|B| - PA)$. $\square$

Lemma 9. Let $q \geq 3$, $b \geq 1$, $N = T_b(q)$, $F(w) = \|Nw\|_1$ and $D = d_b(q)$. Let $P \geq 4$ and $\frac{3}{5} \leq \mu \leq 1$. For all $y \in \{\pm 1\}^{b+1}$ and $\zeta \in [-1, 1]^{b+1}$,

$$\Lambda(y, \zeta) := P F(y - \zeta) + \mu F(\zeta + Py) - (P - \mu) F(\zeta) \leq P(1 + \mu) D,$$

with strict inequality unless $y = \pm s_b$.

Proof. Let $\xi_j(w)$ be the path (1) with $x = q$, $k = b$, and put $\tau_{-1}(w) = \xi_{-1}(w)$, $\tau_j(w) = \xi_{j-1}(w) - \xi_j(w)$ for $0 \leq j < b$. These are linear, and $F(w) = q^b \sum_j |\tau_j(w)|$ by Lemma 5. With $A_j = \tau_j(y)$ and $B_j = \tau_j(\zeta)$,

$$\Lambda/q^b = P(1 + \mu)F(y)/q^b + \sum_j h(A_j, B_j).$$

By Lemma 6 (with $x = q$ and potential μ_j^q), it suffices to show $\sum_j h(A_j, B_j) \leq \sum_{j \in \text{NA}(y)} 2P(1 + \mu)\mu_j^q x_j$, where $x_j = |\xi_j(y)| \in [\frac{q-2}{q}, 1]$, with strict inequality when $\text{NA}(y) \neq \emptyset$. Since $\xi_j(\zeta)$ is a convex combination of entries of ζ , we have $|B_{-1}| \leq 1$ and $|B_j| = \frac{q-1}{q} |\zeta_j - \xi_j(\zeta)| \leq \frac{2(q-1)}{q}$. Also $|A_{-1}| \geq \frac{q-2}{q}$, and $|A_j| = \frac{q-1}{q}(1 + x_j)$ for $j \notin \text{NA}(y)$, $|A_j| = \frac{q-1}{q}(1 - x_j)$ for $j \in \text{NA}(y)$. We use Lemma 8, noting $P \geq 4 > \mu$.

- $j = -1$: $P|A_{-1}| \geq \frac{4(q-2)}{q} \geq \frac{4}{3} > |B_{-1}|$, so $h \leq 0$.
- $j \geq 0$, $j \notin \text{NA}(y)$: $P|A_j| \geq \frac{8(q-1)^2}{q^2} \geq \frac{2(q-1)}{q} \geq |B_j|$, so $h \leq 0$.
- $j \in \text{NA}(y)$: it suffices that $\mu|B_j| < P\beta(x_j)$, where $\beta(t) = \mu \frac{q-1}{q}(1-t) + (1+\mu)\mu_j^q t$. As β is affine it suffices to check the endpoints. For $q = 3$, $\beta(1) \geq \frac{8}{5} \cdot \frac{1}{3}$ and $\beta(\frac{1}{3}) \geq \frac{3}{5} \cdot \frac{4}{9} + \frac{8}{5} \cdot \frac{1}{9} = \frac{4}{9}$, so $P\beta \geq \frac{16}{9} > \frac{4}{3} \geq \mu|B_j|$. For general $q \geq 3$ the same comparison reduces to $P\beta(1) \geq \frac{32}{5} \cdot \frac{q-2}{q} > \frac{2(q-1)}{q}$ and $\frac{22}{5}q^2 - \frac{94}{5}q + \frac{104}{5} > 0$.

Summing over j proves the claim. $\square$

6 The sign-matrix inequality

Theorem 10. Let $p \geq 5$ and $q \geq 3$ be real, $a, b \geq 1$, $M = T_a(p)$ and $N = T_b(q)$. For every $Y \in \{\pm 1\}^{(a+1) \times (b+1)}$,

$$\|MYN^\top\|_1 \leq d_a(p) d_b(q),$$

with equality exactly when $Y = \pm s_a s_b^\top$.

Proof. Let $\bar{y}_0, \dots, \bar{y}_a$ be the rows of Y . Define $\zeta_{a-1} = \bar{y}_a$ and $\zeta_{k-1} = (\zeta_k + (p-1)\bar{y}_k)/p \in [-1, 1]^{b+1}$. Column v of $MYN^\top$ is $Mw^{(v)}$ with $w_i^{(v)} = (N\bar{y}_i)_v$, and by linearity the path of $w^{(v)}$ is $((N\zeta_k)_v)_k$. Applying Lemma 5 with $x = p$ to every column and summing,

$$\|MYN^\top\|_1 = p^a \left(F(\zeta_{-1}) + \sum_{k=0}^{a-1} F(\zeta_{k-1} - \zeta_k) \right), \quad F(w) = \|Nw\|_1.$$

Let μ_k be the potential of Lemma 6 for $x = p$ and put $f_k = F(\zeta_{k-1} - \zeta_k) + \mu_{k-1}F(\zeta_{k-1}) - \mu_k F(\zeta_k)$. Telescoping gives $\|MYN^\top\|_1/p^a = \sum_k f_k + \mu_{a-1}F(\bar{y}_a)$. Since $\zeta_{k-1} - \zeta_k = \frac{p-1}{p}(\bar{y}_k - \zeta_k)$ and $p\mu_k = p-1 - \mu_{k-1}$, we get $pf_k = \Lambda(\bar{y}_k, \zeta_k)$ with $P = p-1 \geq 4$ and $\mu = \mu_{k-1} \in [\frac{p-2}{p}, 1] \subseteq [\frac{3}{5}, 1]$. Lemma 9 gives $f_k \leq c_k d_b(q)$, and Lemma 6 gives $F(\bar{y}_a) \leq d_b(q)$; both are strict unless the row involved is $\pm s_b$, and $\mu_{a-1} > 0$. Hence $\|MYN^\top\|_1 \leq p^a(\sum_k c_k + \mu_{a-1})d_b(q) = d_a(p)d_b(q)$. If equality holds, every row of Y is $\pm s_b$, so $Y = gs_b^\top$ with $g \in \{\pm 1\}^{a+1}$ and $\|MYN^\top\|_1 = \|Mg\|_1 d_b(q)$; by Lemma 6 this equals $d_a(p)d_b(q)$ only for $g = \pm s_a$. $\square$

Remark 11. Since $\|MYN^\top\|_1 = \|NY^\top M^\top\|_1$, the roles of the two factors may be exchanged, so Theorem 10 holds whenever $\min(p, q) \geq 3$ and $\max(p, q) \geq 5$. In particular it contains the sign-matrix inequality [6, Lemma 4.1] used for Theorem 1, and the deduction below then proves Theorem 1 as well. The semidefinite argument of [6] does not extend to $x = 3$ [6, Remark 4.2]; the proof above instead uses the integrality of the signs on both factors.

7 Proof of Theorem 2

Let $a = 2r$, $b = 2s + 1$, $M = T_a(p)$, $N = T_b(3)$, let X be the divisor matrix of $\mathcal{D}$ (so $x_{ab} = 0$), and let $Y = 2X - J$ with J the all-ones matrix. By Proposition 4 and $M\mathbf{1} = p^a e_a$, $N\mathbf{1} = 3^b e_b$,

$$2\mathcal{E}(\text{ICG}_n(\mathcal{D})) = \|MJN^\top + MYN^\top\|_1 = \|n e_a e_b^\top + MYN^\top\|_1 \leq n + d_a(p) d_b(3),$$

by Theorem 10. For the checkerboard set, $Y^* = s_a s_b^\top$, $x_{00}^* = 1$, and $x_{ab}^* = 0$ because $a+b$ is odd, so $\mathcal{D}_{r,s}^*$ is an admissible nonempty set. The (a, b) entry of $MY^*N^\top = (Ms_a)(Ns_b)^\top$ is positive by Lemma 6, so it does not cancel against n , and $2\mathcal{E}(\text{ICG}_n(\mathcal{D}_{r,s}^*)) = n + \|Ms_a\|_1 \|Ns_b\|_1 = n + d_a(p) d_b(3)$. If $\mathcal{E}(\text{ICG}_n(\mathcal{D})) = \mathcal{E}(\text{ICG}_n(\mathcal{D}_{r,s}^*))$, then $\|MYN^\top\|_1 = d_a(p)d_b(3)$, so $Y = \pm s_a s_b^\top$ by Theorem 10; since $y_{ab} = -1 = (s_a s_b^\top)_{ab}$, we get $Y = Y^*$ and $\mathcal{D} = \mathcal{D}_{r,s}^*$. $\square$

8 Exponents of equal parity

Let p, q be distinct odd primes, let $a, b \geq 1$ with $a+b$ even, $n = p^a q^b$, $M = T_a(p)$ and $N = T_b(q)$. For a set $\mathcal{D}$ of proper divisors of n let X and $Y = 2X - J$ be as in Section 7, so that $y_{ab} = -1$, and let $Z = MYN^\top$, with entries Z_{ij} . The last rows of M and N are nonnegative with sums p^a and q^b , so $|Z_{ab}| \leq n$. Writing $t^- = \max\{-t, 0\}$, the computation of Section 7 gives

$$G(Y) := 2\mathcal{E}(\text{ICG}_n(\mathcal{D})) - n = \sum_{(i,j) \neq (a,b)} |Z_{ij}| + Z_{ab} = \|Z\|_1 - 2(Z_{ab})^-. \quad (2)$$

Since $a+b$ is even, the sign matrix $s_a s_b^\top$ has entry $+1$ at (a, b) and violates the proper-divisor condition. The matrix $-s_a s_b^\top$, which corresponds to the set $\{p^i q^j : i+j \text{ odd}\}$, is admissible, and it attains the bound $\|Z\|_1 \leq d_a(p) d_b(q)$ of Theorem 10. However, its entry $Z_{ab} = -(Ms_a)_a (Ns_b)_b = -\delta_a(p) \delta_b(q)$ is negative by Remark 7, and it cancels against n in the argument of Section 7. So the energy bound $\frac{1}{2}[n + d_a(p) d_b(q)]$ is not attained, and the maximiser changes. By (2), the set $\{p^i q^j : i+j \text{ odd}\}$ has energy $\frac{1}{2}[n + d_a(p) d_b(q)] - \delta_a(p) \delta_b(q)$.

Conjecture 12. Let p, q be distinct odd primes, $a, b \geq 1$ with $a + b$ even, and $n = p^a q^b$. The maximal energy of an integral circulant graph of order n is

$$\frac{1}{2}[n + d_a(p) d_b(q)] - \delta_a(p) \delta_b(q),$$

attained exactly by the sets $\{p^i q^j : i + j \text{ odd}\}$ and $\{p^i q^j : i + j \text{ even}\} \setminus \{n\}$.

By the computation above, the first of the two sets always attains the conjectured value. Since $d_1(x) = 3x - 4$ and $\delta_1(x) = x - 2$, Theorem 3 is the case $\min\{a, b\} = 1$ of the conjecture; it is proved in Section 9. The next result is computer-assisted.

Proposition 13. Conjecture 12 holds for the exponent pairs $(a, b) = (2, 2), (2, 4), (4, 2), (2, 6), (6, 2), (2, 8), (8, 2), (3, 3), (3, 5), (5, 3)$ and $(4, 4)$, and all pairs of distinct odd primes p, q .

Proof (computer-assisted). Exchanging the names of the primes transposes X and does not change the energy, and the list of exponent pairs is closed under transposition. As one of two distinct odd primes is at least 5, it therefore suffices to show, for each listed (a, b) and all real $p \geq 5, q \geq 3$, that every $Y \in \{\pm 1\}^{(a+1) \times (b+1)}$ with $y_{ab} = -1$ satisfies $G(Y) \leq \Theta_{ab} := d_a(p) d_b(q) - 2\delta_a(p) \delta_b(q)$, with equality exactly for $Y = -s_a s_b^\top$ and $Y = s_a s_b^\top - 2e_a e_b^\top$, the sign matrices of the two sets in the conjecture; for real p, q , $G(Y)$ is defined by the middle expression in (2).

Substitute $p = 5 + s$ and $q = 3 + t$ with $s, t \geq 0$. Then every entry of Z is a polynomial in s and t with integer coefficients, and so is Θ_{ab} , because $d_k(x)$ and $\delta_k(x) = x^k + 2 \sum_{i < k} (-1)^{k-i} x^i$ have integer coefficients (Remark 7). An entry whose coefficients are all ≥ 0 , or all ≤ 0 , has a known sign on the whole region. Choose a sign ε_{ij} for each of the other entries with $(i, j) \neq (a, b)$, and let L be the sum of Z_{ab} , of the entries of known sign multiplied by their signs, and of the $\varepsilon_{ij} Z_{ij}$. At every point of the region, $G(Y)$ is the largest of these linear forms L . Hence, if for every choice of signs $\Theta_{ab} - L$ either vanishes identically or has nonnegative coefficients and a positive constant term, then $G(Y) \leq \Theta_{ab}$ on the region; equality holds identically if some choice gives $\Theta_{ab} - L = 0$, and $G(Y) < \Theta_{ab}$ everywhere otherwise. For each listed (a, b) , a computer calculation in exact integer arithmetic over all $2^{(a+1)(b+1)-1}$ admissible sign matrices (up to 2^{26}) shows that this condition holds for every Y , and that some choice gives $\Theta_{ab} - L = 0$ exactly when Y is one of the two matrices above. The programs are described in Section 10. $\square$

Together with Theorem 3, Conjecture 12 therefore holds whenever $\min\{a, b\} = 1$ or $(a + 1)(b + 1) \leq 27$. It is open for all other exponent pairs of equal parity; the smallest are $(3, 7), (2, 10)$ and their transposes.

Remark 14. For $n = pq$, Theorem 3 gives the maximal energy $4(p - 1)(q - 1)$, attained exactly by $\{p, q\}$ and $\{1\}$. This can also be read off from known formulas. The graph $\text{ICG}_{pq}(\{1\})$ is the unitary Cayley graph, whose energy $4\varphi(pq)$ was computed by Ilić [3], and $\mathcal{E}(\text{ICG}_{pq}(\{p, q\})) = 4\varphi(pq)$ by [4, Theorem 4.2]. The graphs $\text{ICG}_{pq}(\{p\})$ and $\text{ICG}_{pq}(\{1, p\})$ are p disjoint copies of K_q and the complete q -partite graph with parts of size p (see also [4, Theorem 4.1]); both have energy $2p(q - 1)$, and symmetrically for $\{q\}$ and $\{1, q\}$. Finally $\text{ICG}_{pq}(\{1, p, q\})$ is the complete graph K_{pq} , of energy $2(pq - 1)$. We have not found the maximal energy for $n = pq$ stated explicitly in the literature. Le and Sander [5, Theorem 2.1 and Corollary 2.1] determined the maximum $\tilde{E}_{\max}(n)$ of the energy over *multiplicative* divisor sets and showed $E_{\max}(n) \geq \tilde{E}_{\max}(n)$. For $n = pq$ equality holds, but for $n = pq^3$ it fails: by direct computation, $E_{\max}(5 \cdot 3^3) = 584 > 512 = \tilde{E}_{\max}(5 \cdot 3^3)$ and $E_{\max}(3 \cdot 5^3) = 1632 > 1472 = \tilde{E}_{\max}(3 \cdot 5^3)$, and in both cases the maximisers are the non-multiplicative sets of Theorem 3.

Remark 15. Other sets come close to the maximum. Let $(a, b) = (2, 2)$, $q = 3$ and $\mathcal{D} = \{1, 3, p^2, 3p^2, 9p\}$, so that $Y = s_2 y^\top$ with $y = (1, 1, -1)^\top$ and $Z = (Ms_2)(Ny)^\top$. The path of y is $z_1 = -1, z_0 = \frac{1}{3}, z_{-1} = \frac{7}{9}$, and $\text{NA}(y) = \{0\}$, so Lemma 6 gives $\|Ny\|_1 = d_2(3) - 2 = 23$, and $Z_{22} = 9\delta_2(p)z_{-1} > 0$. By (2), $G(Y) = 23d_2(p)$, while the maximum of G is $25d_2(p) - 10\delta_2(p) =$

$G(Y) + 4(p - 3)$ (Proposition 13). So the energy of $\mathcal{D}$ is smaller than the maximal energy $62p^2 - 90p + 40$ only by $2(p - 3)$.

9 Proof of Theorem 3

Let p, q, m and $n = pq^m$ be as in Theorem 3; the case $n = p^m q$ follows by exchanging the names of the primes. Put $x = q$, $P = x - 1$ and $R = p - 1$. As p and q are distinct odd primes, either $x \geq 5$ and $R \geq 2$, or $x = 3$ and $R \geq 4$; only these inequalities are used. In both cases

$$R + 1 < P(R - 1). \quad (3)$$

Let μ_j and c_j be the potential of Lemma 6 for x , and put $\rho = \mu_{m-1}$, $\hat{c}_j = c_j$ for $0 \leq j < m$, $\hat{c}_m = \rho$ and $\hat{D} = \sum_{j=0}^m \hat{c}_j$. By Lemma 6 and Remark 7, $\hat{D} = d_m(x)/x^m$ and $\rho = \delta_m(x)/x^m$. For $w \in \mathbb{R}^{m+1}$ let $z_j(w)$, $-1 \leq j \leq m-1$, be the path (1) with $k = m$, and let $\nu(w) = \|T_m(x)w\|_1/x^m$. For a sign vector y put $\sigma(y) = \sum_{j \in \text{NA}(y)} \mu_j |z_j(y)|$, so that $\nu(y) = \hat{D} - 2\sigma(y)$ by Lemma 6. For a statement S , $[S]$ is 1 if S holds and 0 otherwise.

For $0 \leq \mu \leq 1$ let h_μ be the function h of Lemma 8 with $P = x - 1$. Besides Lemma 8 we use three values, which follow from the definition and from the case analysis in the proof of Lemma 8: $h_\mu(0, B) = 2\mu|B|$; $h_\mu(A, B) = -2(P - \mu)|B|$ if $AB > 0$ and $|B| \leq |A|$; and $h_\mu(A, B) = 0$ if $AB \leq 0$ and $|B| \leq P|A|$.

Let $\mu_\infty = P/(x + 1)$. As $x\mu_\infty = P - \mu_\infty$, the recursion for μ gives $\mu_j - \mu_\infty = (-1/x)^{j+1}(1 - \mu_\infty)$, where $1 - \mu_\infty > 0$. Hence, for $j \geq 0$,

$$\mu_j > \mu_\infty \text{ if } j \text{ is odd,} \quad \mu_j < \mu_\infty \text{ if } j \text{ is even,} \quad \mu_0 \leq \mu_j \leq \mu_1. \quad (4)$$

In particular $\rho < \mu_\infty$, because $m - 1$ is even.

Lemma 16. *For every $w \in \mathbb{R}^{m+1}$,*

$$\nu(w) = \sum_{j=0}^m \hat{c}_j |w_j| + \frac{1}{x} \sum_{j=0}^{m-1} h_{\mu_{j-1}}(w_j, z_j(w)).$$

Proof. Write $z_j = z_j(w)$. By (1), $z_{j-1} - z_j = \frac{P}{x}(w_j - z_j)$ and $xz_{j-1} = z_j + Pw_j$. Together with $x\mu_j = P - \mu_{j-1}$ this gives, for $0 \leq j < m$,

$$|z_{j-1} - z_j| + \mu_{j-1}|z_{j-1}| - \mu_j|z_j| = \frac{1}{x} [h_{\mu_{j-1}}(w_j, z_j) + P(1 + \mu_{j-1})|w_j|].$$

Summing over j , the potential terms telescope to $\mu_{-1}|z_{-1}| - \mu_{m-1}|z_{m-1}| = |z_{-1}| - \rho|w_m|$, and Lemma 5 gives the claim. $\square$

Lemma 17. *For $y \in \{\pm 1\}^{m+1}$ we have $\nu(y) - |z_{-1}(y)| \leq \hat{D} - \rho$, with equality exactly when $y = \pm s_m$.*

Proof. By Lemma 5, the left side is $\sum_{j < m} |z_{j-1} - z_j|$, where $z_j = z_j(y)$. Let $\mu'_{-1} = 0$ and $\mu'_j = (P - \mu'_{j-1})/x$ for $j \geq 0$, so that $\mu'_j \in (0, P/x]$ for $j \geq 0$. The computation in the proof of Lemma 6 uses only this recursion of the potential, so it gives

$$|z_{j-1} - z_j| + \mu'_{j-1}|z_{j-1}| - \mu'_j|z_j| = c'_j - 2\mu'_j|z_j| \quad [j \in \text{NA}(y)], \quad c'_j = P(1 + \mu'_{j-1})/x.$$

Summing over $j < m$ and using $\mu'_{-1} = 0$ and $|z_{m-1}| = 1$,

$$\sum_{j < m} |z_{j-1} - z_j| = \sum_{j < m} c'_j + \mu'_{m-1} - 2 \sum_{j \in \text{NA}(y)} \mu'_j |z_j|.$$

All terms of the last sum are positive, so the left side is maximal exactly when $\text{NA}(y) = \emptyset$, that is, when $y = \pm s_m$. For $y = s_m$ it equals $\nu(s_m) - |z_{-1}(s_m)| = \hat{D} - \rho$, by Lemma 6 and Remark 7. $\square$

Lemma 18. *Let $y \in \{\pm 1\}^{m+1}$ with $y_0 = y_m$. Then $(3R - 1)\sigma(y) > (R - 1)\rho$.*

Proof. As m is odd and $y_0 = y_m$, $y \neq \pm s_m$, so $\text{NA}(y)$ is nonempty; let $J = \max \text{NA}(y)$. Since y alternates on $J + 1, \dots, m$, the recursion $|z_{j-1}| = (P - |z_j|)/x$ from the proof of Lemma 6 holds for $J < j \leq m - 1$, and starting from $|z_{m-1}| = 1 = \mu_{-1}$ it gives $|z_J(y)| = \mu_{m-2-J}$. Hence $\sigma(y) \geq \mu_J \mu_{m-2-J}$. If $J = m - 1$, this is ρ , and $(3R - 1)\rho > (R - 1)\rho$. Otherwise J and $m - 2 - J$ are nonnegative and have opposite parity, since their sum is odd, so $\sigma(y) > \mu_0 \mu_\infty$ by (4). As $(3R - 1)(x - 2) - (R - 1)x = 2R(x - 3) + 2 > 0$, we have $(3R - 1)\mu_0 > R - 1$, and therefore $(3R - 1)\sigma(y) > (3R - 1)\mu_0 \mu_\infty \geq (R - 1)\mu_\infty \geq (R - 1)\rho$. $\square$

Proof of Theorem 3. Let $\mathcal{D}$ be a set of proper divisors of n , and let u and v be the two rows of its sign matrix Y , so that $v_m = -1$. Since $T_1(p) = \begin{pmatrix} -R & R \\ R & 1 \end{pmatrix}$, the rows of $Z = T_1(p)YT_m(x)^\top$ are $-RT_m(x)w^1$ and $T_m(x)w^2$, where $w^1 = u - v$ and $w^2 = Ru + v$. By the proof of Lemma 5, $Z_{1m} = x^m z_{-1}(w^2)$, so (2) gives $G(Y) = x^m \Phi$ with

$$\Phi = R\nu(w^1) + \nu(w^2) - 2(z_{-1}(w^2))^-.$$

As $d_1(p) = 3R - 1$ and $\delta_1(p) = R - 1$, the claimed maximal energy is $\frac{1}{2}(n + x^m \Theta)$ with $\Theta = (3R - 1)\hat{D} - 2(R - 1)\rho$. We show that $\Phi \leq \Theta$, with equality exactly for $Y = s_1 s_m^\top - 2e_1 e_m^\top$ and $Y = -s_1 s_m^\top$. These are the sign matrices of the two sets in the theorem, which are nonempty sets of proper divisors of n because $1 + m$ is even; so the theorem follows.

Let $\mathcal{A} = \{j : u_j = v_j\}$. For $j \in \mathcal{A}$ we have $w_j^1 = 0$ and $|w_j^2| = R + 1$; for $j \notin \mathcal{A}$ we have $|w_j^1| = 2$ and $|w_j^2| = R - 1$. So $R|w_j^1| + |w_j^2|$ is $R + 1$ on $\mathcal{A}$ and $3R - 1$ off $\mathcal{A}$, and Lemma 16, applied to w^1 and w^2 , gives

$$\Theta - \Phi = 2(R - 1) \left(\sum_{j \in \mathcal{A}} \hat{c}_j - \rho \right) - H + 2(z_{-1}(w^2))^- , \quad (5)$$

where H is the sum of the $2m$ cells $\frac{R}{x} h_{\mu_{j-1}}(w_j^1, z_j(w^1))$ and $\frac{1}{x} h_{\mu_{j-1}}(w_j^2, z_j(w^2))$, $0 \leq j < m$. Recall that $z_j(w)$ is a convex combination of $w_{j+1}, \dots, w_m$.

- (a) For the w^2 -cells, $|z_j(w^2)| \leq R + 1 < P(R - 1) \leq P|w_j^2|$ by (3), so they are ≤ 0 by Lemma 8.
- (b) For the w^1 -cells with $j \notin \mathcal{A}$, $|z_j(w^1)| \leq 2 \leq P|w_j^1|$, so they are ≤ 0 .
- (c) A w^1 -cell with $j \in \mathcal{A}$ equals $\frac{R}{x} h_{\mu_{j-1}}(0, z_j(w^1)) = \frac{2R}{x} \mu_{j-1} |z_j(w^1)|$.

For $j \in \mathcal{A}$ with $j < m$ let

$$s_j = 2(R - 1)c_j - \frac{2R}{x} \mu_{j-1} |z_j(w^1)| \geq \frac{2}{x} [(R - 1)P(1 + \mu) - 2R\mu], \quad \mu = \mu_{j-1},$$

where we used $|z_j(w^1)| \leq 2$. The bracket is affine in μ , and it equals $(R - 1)P > 0$ at $\mu = 0$ and $2[R(P - 1) - P] > 0$ at $\mu = 1$; hence $s_j > 0$. Let $S = \sum_{j \in \mathcal{A}, j < m} s_j$, let $C \geq 0$ be minus the sum of the cells in (a) and (b), and let $K = 2(z_{-1}(w^2))^- \geq 0$. Then (5) reads

$$\Theta - \Phi = S + C + K - 2(R - 1)\rho [m \notin \mathcal{A}]. \quad (6)$$

Moreover $|z_0(w^2)| \leq R + 1 < P|w_0^2|$, so $z_{-1}(w^2) = (z_0(w^2) + Pw_0^2)/x$ has the sign of $w_0^2 = Ru_0 + v_0$, which is the sign of u_0 . Thus $K = 0$ if and only if $u_0 = 1$.

Case 1: $m \in \mathcal{A}$. Then $\Theta - \Phi = S + C + K \geq 0$. Suppose that equality holds. Then $S = 0$, so $\mathcal{A} = \{m\}$; $K = 0$, so $u_0 = 1$; and $C = 0$. Now $w_j^1 = 2u_j$ for $j < m$ and $w_m^1 = 0$. For $j \leq m - 2$ we have $|Pw_{j+1}^1| = 2P > 2 \geq |z_{j+1}(w^1)|$, so $z_j(w^1) = (z_{j+1}(w^1) + Pw_{j+1}^1)/x$ has the sign of u_{j+1} , and $|z_j(w^1)| \leq 2 = |w_j^1|$. If $u_j = u_{j+1}$ for some $j \leq m - 2$, the w^1 -cell at j is

therefore $-\frac{2R}{x}(P - \mu_{j-1})|z_j(w^1)| < 0$, contradicting $C = 0$. So u alternates on $0, \dots, m-1$ and $u_0 = 1$; together with $u_m = v_m = -1$ and $v_j = -u_j$ for $j < m$, this means $Y = s_1 s_m^\top - 2e_1 e_m^\top$, as m is odd. Conversely, for this Y we have $\mathcal{A} = \{m\}$, $u = s_m$ and $S = K = 0$. The w^1 -cells vanish: for $j \leq m-2$, $w_j^1 = 2u_j$ and $z_j(w^1)$ have opposite signs and $|z_j(w^1)| \leq P|w_j^1|$, and $z_{m-1}(w^1) = w_m^1 = 0$. The w^2 -cells vanish too: w_i^2 has the sign of u_i for every i ; by (3), $z_j(w^2) = (z_{j+1}(w^2) + Pw_{j+1}^2)/x$ has the sign of w_{j+1}^2 for $j \leq m-2$, and $z_{m-1}(w^2) = w_m^2$; since u alternates, w_j^2 and $z_j(w^2)$ have opposite signs, and $|z_j(w^2)| \leq P|w_j^2|$ by (a). Hence $\Phi = \Theta$.

Case 2: $m \notin \mathcal{A} \neq \emptyset$. Then $u_m = 1$ and $\mathcal{A} \subseteq \{0, \dots, m-1\}$. By (6) it suffices to find $k \in \mathcal{A}$ with $s_k > 2(R-1)\rho$. Since $\rho < \mu_\infty = P/(P+2)$, the lower bound for s_k shows that this holds if

$$(R-1)P + \mu(P+2)[R(P-2) - P] \geq 0, \quad \mu = \mu_{k-1}. \quad (7)$$

If $x \geq 5$, then $P \geq 4$ and $R(P-2) - P \geq P-4 \geq 0$, so (7) holds for every $k \in \mathcal{A}$. If $x = 3$, then (7) reads $2(R-1) \geq 8\mu$. If some $k \in \mathcal{A}$ has $k \geq 1$, then $\mu_{k-1} \leq \mu_1 = \frac{5}{9}$ by (4), and $2(R-1) \geq 6 > \frac{40}{9}$. It remains to treat $x = 3$ and $\mathcal{A} = \{0\}$. Here $\mu_\infty = \frac{1}{2}$ and $c_0 = \frac{4}{3}$, and since $z_0(w)$ depends only on $w_1, \dots, w_m$, we have $z_0(w^1) = 2z_0(u)$ and $s_0 = \frac{2}{3}[4(R-1) - 2R|z_0(u)|]$. If $m = 1$, then $|z_0(u)| = 1$, $\rho = \mu_0 = \frac{1}{3}$ and $s_0 = \frac{2}{3}(2R-4) > \frac{2}{3}(R-1) = 2(R-1)\rho$. Let $m \geq 3$. If $|z_0(u)| \leq \frac{5(R-1)}{4R}$, then $s_0 \geq R-1 > 2(R-1)\rho$. Otherwise $|z_0(u)| > \frac{15}{16}$. If $u_1 \neq u_2$, then $u_1 = -\text{sgn } z_1(u)$ by Lemma 6, and $|z_0(u)| = (2 - |z_1(u)|)/3 \leq \frac{5}{9}$; hence $u_1 = u_2$. Then $1 \notin \mathcal{A}$, and the arguments $w_1^1 = 2u_1$ and $z_1(w^1) = 2z_1(u)$ of the w^1 -cell at $j = 1$ have the same sign, with $|z_1(w^1)| \leq |w_1^1|$. This cell equals $-\frac{2R}{3}(P - \mu_0)|z_1(w^1)| = -\frac{20R}{9}|z_1(u)| \leq -\frac{20R}{27}$, so $C \geq \frac{20R}{27}$. With $s_0 \geq \frac{2}{3}(2R-4)$ and $2(R-1)\rho < R-1$,

$$\Theta - \Phi \geq \frac{2}{3}(2R-4) + \frac{20R}{27} - 2(R-1)\rho > \frac{56R-72}{27} - (R-1) = \frac{29R-45}{27} > 0.$$

Case 3: $\mathcal{A} = \emptyset$, that is, $v = -u$. Then $u_m = 1$, $w^1 = 2u$, $w^2 = (R-1)u$ and $\Phi = (3R-1)\nu(u) - 2(R-1)(z_{-1}(u))^-$. If $u_0 = 1$, then $z_{-1}(u) > 0$ by Lemma 6, and $\Phi = (3R-1)(\hat{D} - 2\sigma(u)) < \Theta$ by Lemma 18, since $u_0 = u_m$. If $u_0 = -1$, then $(z_{-1}(u))^- = |z_{-1}(u)|$ and

$$\Phi = (R+1)\nu(u) + 2(R-1)(\nu(u) - |z_{-1}(u)|) \leq (R+1)\hat{D} + 2(R-1)(\hat{D} - \rho) = \Theta$$

by Lemmas 6 and 17, with equality exactly when $u = \pm s_m$, that is, $u = -s_m$. This is $Y = -s_1 s_m^\top$.

The three cases cover all Y , which proves the theorem. For $m = 1$ the value is $4(p-1)(q-1)$, since $d_1(q) = 3q-4$ and $\delta_1(q) = q-2$. $\square$

10 Verification

The proofs above are complete, except that the proof of Proposition 13 rests on the computer calculation described in the paragraph on certificates below. The following computations are independent cross-checks.

- An exhaustive search over all divisor sets, in exact integer arithmetic, confirms Theorem 2, including the value of the maximum, in the following cases:
 - exponent pairs $(4, 1)$, $(6, 1)$, $(8, 1)$, $(2, 5)$, $(4, 3)$, for the primes $5 \leq p \leq 31$;
 - $(10, 1)$, $(12, 1)$, $(2, 7)$, $(6, 3)$, for $p \in \{5, 7, 11, 13, 17\}$;
 - $(14, 1)$, $(4, 5)$, $(2, 9)$, for $p \in \{5, 7, 11\}$ (the largest case has 2^{29} divisor sets).
- Theorem 10 was checked in exact arithmetic over all sign matrices of every shape with $(a+1)(b+1) \leq 25$, for $(p, q) = (5, 3)$, $(7, 3)$ and $(5, 4)$.

- Each lemma was tested separately in exact arithmetic: by adversarial searches for small b , and by structured random tests of Lemma 9 up to $b = 20$.
- Proposition 4 was compared, in floating-point arithmetic, with spectra of the actual circulant matrices. The fast Fourier transform was used for all divisor sets of the orders $n = 75, 147, 363, 675, 1323, 1875$ and 6075 . Dense diagonalisation was used for all divisor sets when $n \leq 675$ and for samples of them when $n = 1323, 1875$ and 6075 .
- Theorem 3 was confirmed by exhaustive searches over all divisor sets: in exact arithmetic, with the eigenvalues computed from Ramanujan sums and independently of Proposition 4, for 35 orders pq^m with $m \in \{1, 3, 5, 7\}$ up to $n = 234375$; and with floating-point spectra from the fast Fourier transform for ten orders up to $n = 9375$.
- The inequality $\Phi \leq \Theta$ of Section 9, with its equality cases, was checked in exact arithmetic over all pairs of sign vectors for $m \leq 11$ at several parameter pairs (x, R) . Each intermediate claim of its proof was checked separately for $m \leq 7$, including non-integer parameters.
- An exhaustive search over all divisor sets confirms Conjecture 12 for ten ordered pairs of odd primes and each of the exponent pairs $(1, 1), (2, 2), (1, 3), (3, 1), (3, 3), (2, 4), (4, 2), (4, 4), (1, 5)$ and $(3, 5)$. These cases are now covered by Theorem 3 and Proposition 13.

Certificates for Proposition 13. The certificate is implemented in Python, in exact integer arithmetic, in two versions that share the routines building the polynomial matrices. A vectorised version uses 64-bit integers, with an asserted bound that excludes overflow; it first applies the cheaper sufficient test in which the absolute value of each entry of undetermined sign is bounded by the polynomial with the absolute values of its coefficients, and branches over signs only when this test fails. A pure-Python version branches over the signs of the undetermined entries for every sign matrix. The vectorised version certified all eleven exponent pairs; the pure-Python version gave the same result for $(2, 2), (2, 4), (4, 2)$ and $(3, 3)$. An independent re-implementation in SymPy, which builds $T_k(x)$ from the Ramanujan-sum formula and recomputes the target as $G(-s_a s_b^T)$, checking it against the closed form, confirmed $(2, 2), (2, 4), (4, 2)$ and $(3, 3)$. The seven larger exponent pairs were run only with the vectorised version. The polynomial matrices were compared with the exact matrices $T_k(x)$, $k \leq 8$, at sample points. As negative controls, the certificate is rejected when the target is lowered by 1, and, for $(a, b) = (1, 1)$ and $(2, 2)$, when the region is enlarged to $p, q \geq 3$: at $p = q = 3$ further sign matrices attain the target. Proposition 13 is not formalised.

Formal verification. Theorem 10, and Theorems 2 and 3 with $d_k(x)$ taken as $\|T_k(x)s_k\|_1$ and $\delta_k(x)$ as $(T_k(x)s_k)_k$, are verified in Lean 4 (version 4.33.1, Mathlib commit 0df444a3). The development starts from the definition of the graph and of its energy:

- the adjacency matrix of $\text{ICG}_n(\mathcal{D})$ has entry 1 exactly when $\gcd((j - i) \bmod n, n) \in \mathcal{D}$;
- the energy is the sum of the absolute values of its eigenvalues, as given by Mathlib's spectral theorem.

It proves:

- Lemmas 5, 6, 8 and 9, and Theorem 10, for real $p \geq 5$, $q \geq 3$ and all $a, b \geq 1$, together with the exchanged form of Remark 11;
- Proposition 4 for all distinct primes p, q and all exponents, from the eigenvalues of circulant matrices;

- the conclusion of Theorem 2 and the value of the maximal energy, with $d_k(x)$ defined as $\|T_k(x)s_k\|_1$;
- the conclusion of Theorem 1 for all pairs of distinct odd primes, which includes Theorem 2 and the case $q \geq 5$;
- Theorem 3, as the theorem `ICGEQUALParity.corollaryA`. For distinct primes $p, q \geq 3$ and odd m , with $n = pq^m$, $\mathcal{D}^- = \{p^i q^j : i + j \text{ odd}\}$ and $\mathcal{D}^+ = \{p^i q^j : i + j \text{ even}\}$, it states that $\mathcal{D}^-$ and $\mathcal{D}^+ \setminus \{n\}$ are sets of proper divisors of n ; that $2\mathcal{E}(\text{ICG}_n(\mathcal{D}^-)) = n + (3p - 4)\|T_m(q)s_m\|_1 - 2(p - 2)(T_m(q)s_m)_m$; that $\mathcal{E}(\text{ICG}_n(\mathcal{D}^+ \setminus \{n\})) = \mathcal{E}(\text{ICG}_n(\mathcal{D}^-))$; and that every set $\mathcal{D}$ of proper divisors of n satisfies $\mathcal{E}(\text{ICG}_n(\mathcal{D})) \leq \mathcal{E}(\text{ICG}_n(\mathcal{D}^-))$, with equality only if $\mathcal{D} = \mathcal{D}^-$ or $\mathcal{D} = \mathcal{D}^+ \setminus \{n\}$. The case $n = p^m q$ is the same statement with the primes exchanged. The inequality $\Phi \leq \Theta$ of Section 9, strict except for the two extremal sign matrices, is proved for all real x, R with $x \geq 5$, $R \geq 2$ or $x = 3$, $R \geq 4$.

The closed forms of $d_k(x)$ and $\delta_k(x)$ in Remark 7 are proved above but not formalised; in the Lean statements, $d_m(q)$ and $\delta_m(q)$ appear as $\|T_m(q)s_m\|_1$ and $(T_m(q)s_m)_m$. The whole development was rebuilt from a clean directory; the files for Theorem 3 were rebuilt in the same way, together with all files they import. The recorded axiom audits of the main results (the command `#print axioms`) report only `propext`, `Classical.choice` and `Quot.sound`; no native evaluation is used. The code and logs are available at [1].

Tool and computational resource disclosure

This work was carried out between 25 and 26 September 2026 in a workflow directed by the author, using the AI coding agents Anthropic Claude Code (model Claude Opus 5.5) and OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`). The author chose the research programme and directed the agents.

- A Claude Code agent found the proof of Theorem 10 and of Theorem 2, formulated Conjecture 12, and wrote the verification programs.
- Another Claude Code agent wrote the Lean formalisation.
- The proof was checked line by line by a second Claude Code agent and by Codex, and the agents searched the literature.
- The results on exponents of equal parity, Theorem 3 and Proposition 13, were found by a Claude Code research agent, and Theorem 3 was checked by an independent Claude referee agent.

Claude Code drafted this text. The programs and logs are available at [1]. The author is not a professional mathematician. The author has read the paper and, using a side-by-side table, compared the statements of the Lean theorems with the theorems stated here. The correctness of Theorems 2, 10 and 3 rests on the written proofs and on the Lean formalisation described in Section 10; Proposition 13 rests on the computer calculation described there; the other independent checks are additional cross-checks. In addition, the arguments for Theorems 2 and 10 were checked by AI agents of two different model families, the argument for Theorem 3 was checked by an independent Claude agent, and the formal statements were reviewed by an agent that did not write them. AI systems are not authors of this paper; the author takes full responsibility for its content.

References

- [1] H. Dong, *ai4math-results*, repository, <https://github.com/Horace-Maxwell/ai4math-results>; archived at Zenodo, <https://doi.org/10.5281/zenodo.22970254>.
- [2] I. Gutman, The energy of a graph, *Ber. Math.-Statist. Sect. Forsch. Graz* **103** (1978), 1–22.
- [3] A. Ilić, The energy of unitary Cayley graphs, *Linear Algebra Appl.* **431** (2009), 1881–1889.
- [4] A. Ilić and M. Bašić, New results on the energy of integral circulant graphs, *Appl. Math. Comput.* **218** (2011), 3470–3482.
- [5] T. A. Le and J. W. Sander, Extremal energies of integral circulant graphs via multiplicity, *Linear Algebra Appl.* **437** (2012), 1408–1421, <https://doi.org/10.1016/j.laa.2012.04.012>.
- [6] J. Jiang and C. Yang, Energy maximisation for integral circulant graphs with opposite-parity exponents, arXiv:2608.29523v1, 30 August 2026.
- [7] S. Park, *Exact energy maximisation for integral circulant graphs of order p^2q^3* , manuscript posted 21 August 2026, <https://github.com/coshaman/papers/tree/2f730393e3d5bed4d9641d7d79b1bdc57fbff05d>.
- [8] D. G. Roldán, Graph energy maximisation for integral circulant graphs of order $n = p^2q^3$, arXiv:2604.09491v1, 10 April 2026.
- [9] J. W. Sander and T. Sander, The exact maximal energy of integral circulant graphs with prime power order, *Contrib. Discrete Math.* **8** (2013), 19–40.
- [10] J. A. Schreib, *The Roldan universality conjecture*, repository, 5 September 2026, <https://github.com/jamesschreib/roldan-universality-conjecture/tree/95bb002688c65948a15cc9380e8e1f08f8bd6801>.
- [11] W. So, Integral circulant graphs, *Discrete Math.* **306** (2006), 153–158.
- [12] The Omega Institute, *Roldán integral-circulant energy: exceptional $q = 3$ case*, issue #8338 of the `trureturing` repository, opened 16 September 2026, <https://github.com/the-omega-institute/trureturing/issues/8338>.